

## Privacy Concerns

**Affect**

**Many**

**Aspects**

**of**

An individual's

**life**

Commerce

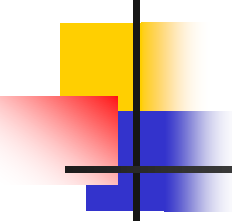
Health Care

Education

Work

Recreation

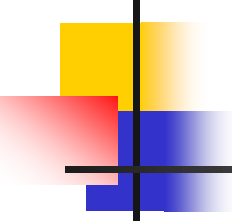
Our day-to-day lives



# What is Personal Privacy

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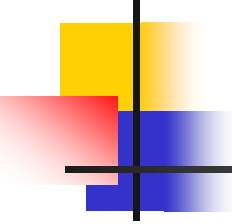
- Privacy is a concept that is difficult to define.
- We sometimes speak of an individual's privacy as something that can be:
  - lost,
  - diminished,
  - intruded upon,
  - invaded,
  - violated,
  - breached.



# What is Privacy (continued)?

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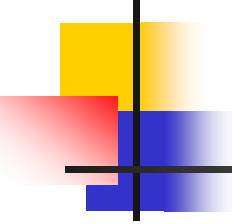
- Privacy is sometimes viewed in terms of something that can be *diminished* (i.e., as a repository of personal information that can be eroded gradually) or *lost* altogether.
- Privacy is sometimes also construed in terms of the metaphor of a (spatial) zone that can be *intruded* upon or *invaded*.
- Privacy is also sometimes analyzed in terms of concerns affecting the confidentiality of information, which can be *breached* or *violated*.



# Classic Theories of Privacy

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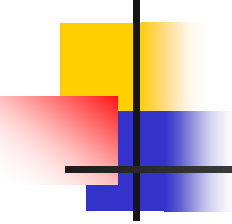
- Traditional (or classic) privacy theories have tended to view privacy in connection with notions such as:
  - non-intrusion (into one's space),
  - non-interference (with one's decisions),
  - having control over/restricting access to one's personal information.



# Non-intrusion Theories of Privacy

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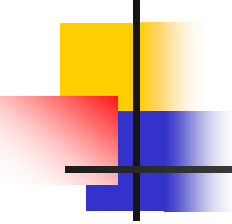
- Non-intrusion theories view privacy as either:
  - being let alone,
  - being free from government intrusion (into one's physical space).
- This view is also sometimes referred to as *accessibility privacy* (DeCew, 1997).



# Non-interference Theories of Privacy

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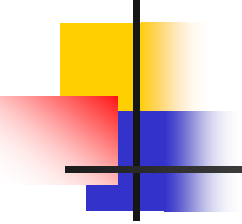
- Non-interference theories view privacy in terms of freedom from interference in making decisions.
- This perspective emerged in the 1960s, following the *Griswold v. Connecticut* (U.S. Supreme Court) case in 1965.
- This view of privacy is also sometimes referred to as *decisional privacy*.



# The Control and Limited Access Theories of Informational Privacy

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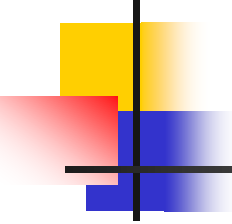
- *Informational privacy* is concerned with protecting personal information in computer databases.
- Most people wish to have some *control* over their personal information.
- In some cases, “privacy zones” have been set up either to restrict or limit *access* to one’s personal data.



# Table 5-1: Three Views of Privacy

|                              |                                                                                                                                     |
|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| <i>Accessibility Privacy</i> | Privacy is defined in terms of one's physically "being let alone," or freedom from intrusion into one's physical space.             |
| <i>Decisional Privacy</i>    | Privacy is defined in terms of freedom from interference in one's choices and decisions.                                            |
| <i>Informational Privacy</i> | Privacy is defined as control over the flow of one's personal information, including the transfer and exchange of that information. |

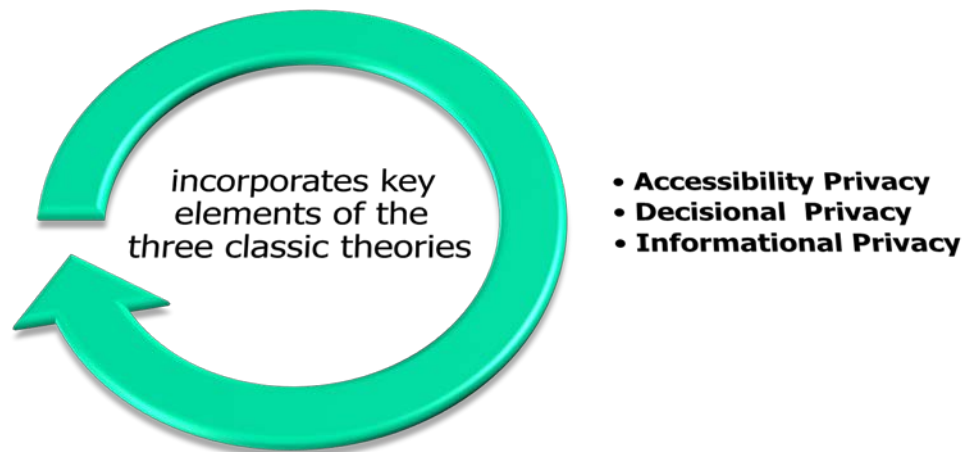


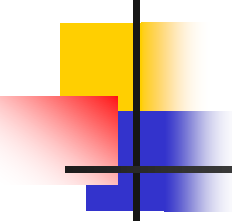


# A Comprehensive Account of Privacy

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- Moor (2004) has articulated a privacy theory that incorporates key elements of the three classic theories



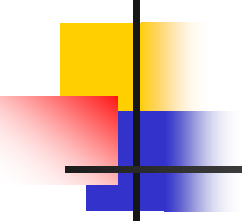


# Moor's Comprehensive Theory of Privacy

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- According to Moor:

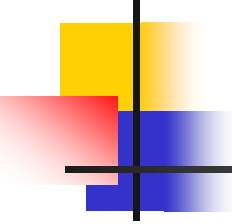
“an individual has privacy in a *situation* if in that particular situation the individual is *protected from intrusion, interference, and information access* by others.”



# Moor's Theory of Privacy (continued)

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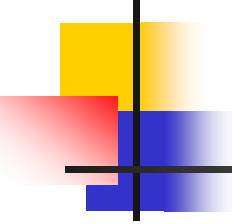
- A key element in Moor's definition is his notion of a *situation*, which can apply to a range of contexts or “zones.”
- For Moor, a situation can be an “activity,” a “relationship,” or the “storage and access of information” in a computer or on the Internet.



# Moor's Privacy Theory (continued)

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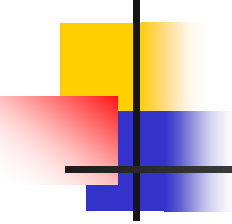
- Moor also distinguishes between “naturally private” and “normatively private” situations required for having:
  - a) natural privacy (in a descriptive sense);
  - b) a right to privacy (in a normative sense).



# Applying Moor's Natural vs. Normative Privacy Distinction

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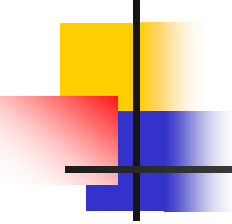
- Using Moor's natural/normative privacy distinction, we can further differentiate between a:
  - *loss of privacy,*
  - *violation of privacy.*



# Descriptively Private vs. Normatively Private Situations

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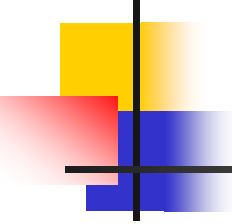
- Tom walks into the computer lab (when no one else is around) and sees Mary in the lab.
  - In this natural/descriptively private situation, Mary's privacy is lost but not violated.
- Tom peeps through the keyhole of Mary's apartment door and sees Mary typing at her computer.
  - In this normatively private situation, Mary's privacy is not only lost but is also violated.



# Nissenbaum's Theory of Privacy as “Contextual Integrity”

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- Nissenbaum's privacy framework requires that the processes used in gathering and disseminating information are
  - a) “appropriate to a particular context”
  - b) comply with norms that govern the flow of personal information in a given context.

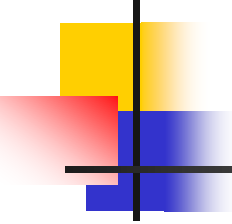


## Nissenbaum's Theory (Continued)

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- Nissenbaum (2004a) refers to these two types of informational norms as:
  - norms of appropriateness,
  - norms of distribution.

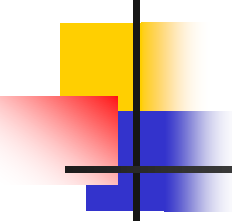




# Nissenbaum's Theory (Continued)

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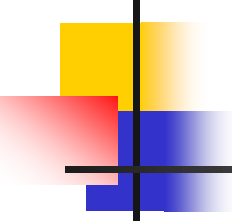
- *Norms of appropriateness* determine whether a given type of personal information is either appropriate or inappropriate to divulge within a particular *context*.
- *Norms of distribution* restrict or limit the flow of information within and across *contexts*.
- When either norm is “breached,” a violation of privacy occurs.
- Conversely, the *contextual integrity* of the flow of personal information is maintained when both kinds of norms are “respected”



# Nissenbaum's Theory (Continued)

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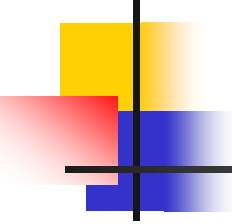
- Like Moor's privacy model, Nissenbaum's theory demonstrates why we must always focus on the *context in which information flows, not the nature of the information itself*, in determining whether normative protection is needed.
- Review Scenario 5-3 (in the textbook) on Professor Robert's seminar, which illustrates the notion of "contextual integrity."



# Can Privacy Be Preserved in the Digital Era?

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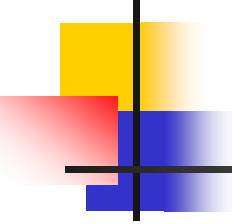
- In 1999, Scott McNealy, CEO of Sun Microsystems, uttered his now famous remark to a group of reporters: *You have zero privacy anyway. Get over it.*
- Froomkin (2000), Garfinkel (2000), and others have expressed concerns about the “death of privacy.”
- But some believe that not all has yet been lost in the battle over privacy.
- For example, some privacy advocates staunchly believe that we should be vigilant about retaining and safeguarding what little privacy we may still have.



# What Kind of Value is Privacy?

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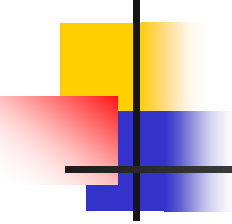
- Three distinct questions can be distinguished with respect to privacy as a *value*:
  1. Is privacy an *intrinsic* value, or is it an *instrumental* value?
  2. Is privacy *universally* valued, or is it valued mainly in Western industrialized societies (where greater importance is placed on the individual than on the broader community?)
  3. Is privacy an important *social* value (as well as an individual value)?



# Is Privacy an Intrinsic Value or an Instrumental Value?

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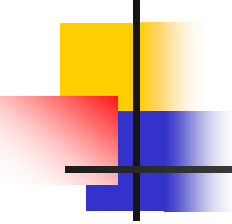
- Is privacy something that is valued for its own sake?
  - In other words, is it an *intrinsic value*?
- Or, is privacy valued as a means to some further end?
  - Is it merely an *instrumental value*?



# Is Privacy an Intrinsic or an Instrumental Value (Continued)?

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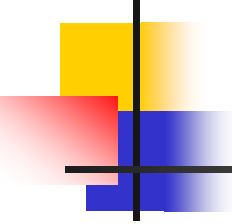
- Privacy does not seem to be valued for its own sake, and thus does not appear to have intrinsic worth.
- But privacy also seems to be more than merely an instrumental value because it is *necessary* (rather than merely *contingent*) for achieving important human ends (Fried, 1990).



# Is Privacy an Intrinsic or an Instrumental Value (Continued)?

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- Fried notes that privacy is necessary for important human ends such as *trust* and *friendship*.
- Moor views privacy as an expression of a “core value” – viz., *security*, which is essential for human flourishing.

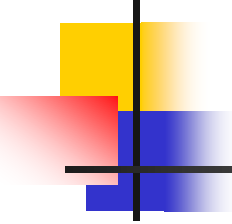


# Privacy as a Universal Value

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- Privacy has at least some importance in all societies, but it is not valued the same in all cultures.
- For example, privacy tends to be less valued in many non-Western nations, as well as in many rural societies in Western nations.
- Privacy also tends to be less valued in some democratic societies where national security and safety are considered more important than individual privacy (e.g., as in Israel).

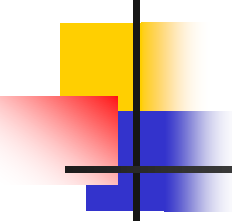




# Privacy as an Important Social Value

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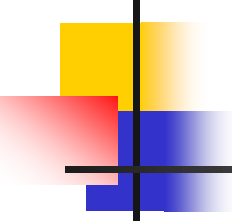
- Priscilla Regan (1995) notes that we tend to underestimate the importance of privacy as an important *social value* (as well as an individual value).
- Regan believes that if we frame the privacy debate in terms of privacy as a social value (essential for democracy), as opposed to an individual good, the importance of privacy is better understood.



# Cybertechnology-related Techniques that Threaten Privacy

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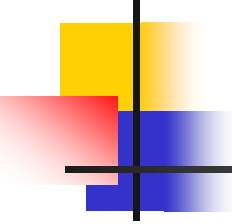
- We examine three techniques that threaten privacy:
  - 1) *data-gathering* techniques used to collect and record personal information, often without the knowledge and consent of users.
  - 2) *data-exchanging* techniques used to transfer and exchange personal data across and between computer databases, typically without the knowledge and consent of users.
  - 3) *data-mining* techniques used to search for patterns implicit in large databases in order to generate consumer profiles based on behavioral patterns discovered in certain groups.



# Cybertechnology Techniques Used to Gather Personal Data

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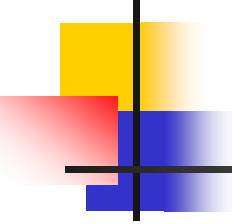
- Personal data has been gathered at least since Roman times (census data).
- Roger Clarke uses the term *dataveillance* to capture two techniques made possible by cybertechnology:
  - a) surveillance (data-monitoring),
  - b) data-recording.



# Internet Cookies as a Surveillance Technique

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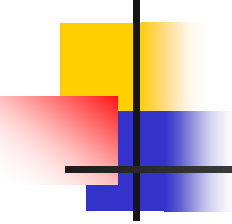
- “Cookies” are files that Web sites send to and retrieve from the computers of Web users.
- Cookies technology enables Web site owners to collect data about those who access their sites.
- With cookies, information about one’s online browsing preferences can be “captured” whenever a person visits a Web site.



# Cookies (Continued)

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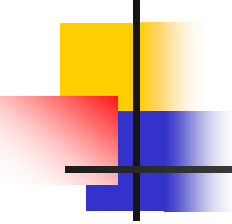
- The data recorded via cookies is stored on a file placed on the hard drive of the user's computer system.
- The information can then be retrieved from the user's system and resubmitted to a Web site the next time the user accesses that site.
- The exchange of data typically occurs without a user's knowledge and consent.



# Can the Use of Cookies be Defended?

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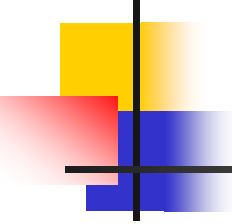
- Many proprietors of Web sites that use cookies maintain that they are performing a service for repeat users of their sites by customizing a user's means of information retrieval.
- For example,, some point out that, because of cookies, they are able to provide a user with a list of preferences for future visits to that Web site.



# Arguments Against Using Cookies

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- Some privacy advocates argue that activities involving the monitoring and recording an individual's activities while visiting a Web site violates privacy.
- Some also worry that information gathered about a user via cookies can eventually be acquired by or sold to online advertising agencies.

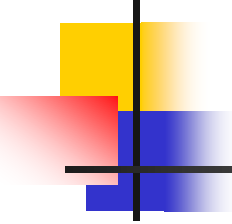


# RFID Technology as a Surveillance Technique

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- RFID (Radio Frequency IDentification) consists of a *tag* (microchip) and a *reader*.
  - The tag has an *electronic circuit*, which stores data, and *antenna* that broadcasts data by radio waves in response to a signal from a reader.
  - The reader contains an *antenna* that receives the radio signal, and *demodulator* that transforms the analog radio into suitable data for any computer processing that will be done (Lockton and Rosenberg, 2005).

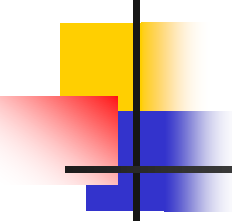




# RFID Technology (Continued)

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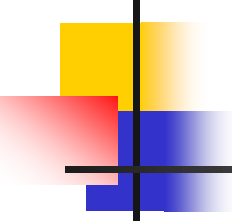
- RFID transponders in the form of “smart labels” make it much easier to track inventory and protect goods from theft or imitation.
- RFID technology also poses a significant threat to individual privacy.
- Critics worry about the accumulation of RFID transaction data by RFID owners and how that data will be used in the future.



# RFID Technology (Continued)

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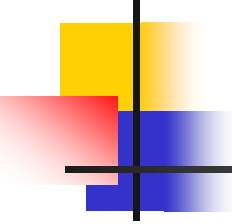
- Garfinkel (2004) notes that roughly 40 million Americans carry some form of RFID device every day.
- Privacy advocates note that RFID technology has been included in chips embedded in humans, which enables them to be tracked.



# RFID Technology (Continued)

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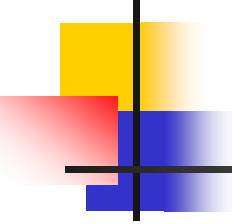
- Like Internet cookies (and other online data gathering and surveillance techniques), RFID threatens individual privacy.
- Unlike cookies, which track a user's habits while visiting Web sites, RFID technology can track an individual's location in the off-line world.
- RFID technology also introduces concerns involving "locational privacy"



# Cybertechnology and Government Surveillance

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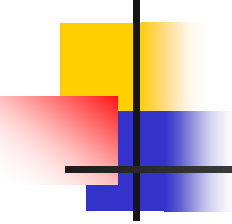
- As of 2005, cell phone companies are required by the FCC to install a GPS (Global Positioning System) locator chip in all new cell phones.
- This technology, which assists 911 operators, enables the location of a cell phone user to be tracked within 100 meters.
- Privacy advocates worry that this information can also be used by the government to spy on individuals.



# Computerized *Merging* Techniques

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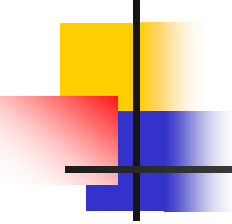
- *Computer merging* is a technique of extracting information from two or more unrelated databases and incorporating it into a composite file.
- Computer merging occurs whenever two or more disparate pieces of information contained in separate databases are combined.



# Computer Merging (Continued)

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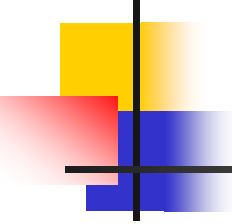
- Imagine a situation in which you voluntarily provide information about yourself to three different organizations – i.e., you give information about your:
  - 1) income and credit history to a lending institution in order to secure a loan;
  - 2) age and medical history to an insurance company to purchase life insurance;
  - 3) views on certain social issues to a political organization you wish to join.



# Computer Merging (Continued)

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- Each organization has a legitimate need for information to make decisions about you; for example:
  - insurance companies have a legitimate need to know about your age and medical history before agreeing to sell you life insurance;
  - lending institutions have a legitimate need to know information about your income and credit history before agreeing to lend you money to purchase a house or a car.

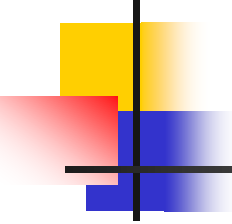


# Computer Merging (Continued)

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- Suppose that information about you in the insurance company's database is merged with information about you in the bank's database or in the political organization's database.
- When you gave certain information about yourself to three different organizations, you authorized each organization to have specific information about you.
- However, it does not follow that you thereby authorized any one organization to have some combination of that information.

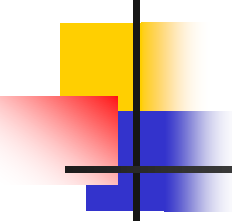




## Computer Merging (Continued)

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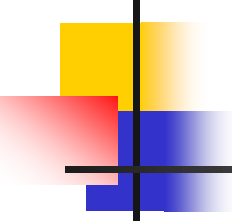
- Review Scenario 5-4 (in the textbook) involving Double-Click (an online advertising company that attempted to purchase Abacus, Inc., an off-line database company).
- If it had succeeded in acquiring Abacus, Double-Click would have been able to merge on- and off-line records.



# Computer *Matching*

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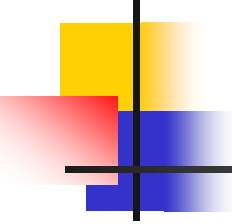
- *Computer matching* is a variation of computer merging.
- Matching is a technique that cross-checks information in two or more databases that are typically unrelated to produce "matching records" or "hits."



# Computer Matching (Continued)

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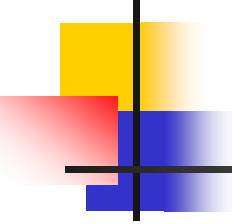
- In practices involving federal and state government organizations, computerized matching has been used by various agencies and departments to identify:
  - *potential* law violators;
  - individuals who have *actually* broken the law or who are suspected of having broken the law (welfare cheats, deadbeat parents, etc.).



# Computer Matching (Continued)

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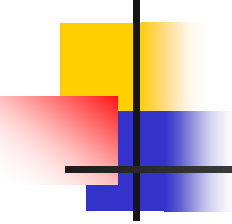
- Income tax records could be matched against state motor vehicle registration records (looking for individuals reporting low incomes but owning expensive automobiles).
- Consider an analogy in physical space where your mail is matched (and opened) by authorities to catch criminals suspected of communicating with your neighbors.



# Computer Matching (Continued)

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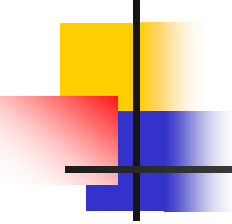
- Some who defend matching argue:  
*If you have nothing to hide, you have nothing to worry about.*
- Others use the following kind of argument:
  1. Privacy is a legal right.
  2. Legal rights are not absolute.
  3. When one violates the law (i.e., commits a crime), one forfeits one's legal rights.
  4. Therefore, criminals have forfeited their right to privacy.



# Computer Matching (Continued)

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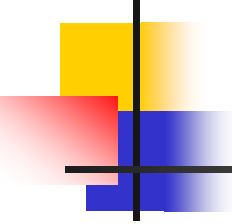
- Review Scenario 5-5 (in the text) involving Super Bowl XXXV (in 2001), where a facial-recognition technology was used to scan the faces of individuals entering the stadium.
- The digitized facial images were instantly matched against images contained in a centralized database of suspected criminals and terrorists.
- This practice was criticized by many civil-liberties proponents at that time (January 2001).



# Data Mining

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- Data mining involves the indirect gathering of personal information via an analysis of implicit patterns discoverable in data.
- Data-mining activities can generate new and sometimes non-obvious classifications or categories.
- Individuals whose data is mined could become identified with or linked to certain newly created groups that they might never have imagined to exist.

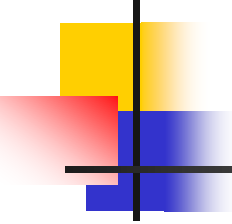


# Data Mining (Continued)

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- Current privacy laws offer individuals little-to-no protection for how personal information that is acquired through data-mining activities is subsequently used.
- Yet, important decisions can be made about individuals based on the patterns found in the personal data that has been “mined.”
- Some uses of data-mining technology raise special concerns for personal privacy.

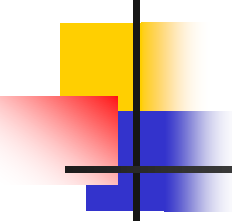




# Data Mining (Continued)

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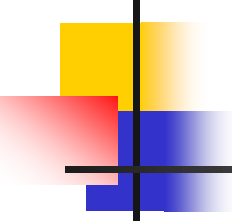
- Why is mining personal data controversial?
- Unlike personal data that resides in explicit records in databases, information acquired about persons via data mining is often derived from implicit patterns in the data.
- The patterns can suggest "new" facts, relationships, or associations about that person, such as that person's membership in a newly "discovered" category or group.



# Data Mining (Continued)

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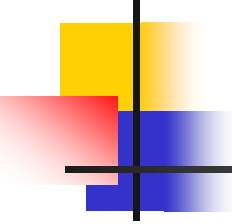
- Much personal data collected and used in data-mining applications is generally considered to be information that is neither confidential nor intimate.
- So, there is a tendency to presume that personal information generated by or acquired via data mining techniques must by default be *public* data.



# Data Mining (Continued)

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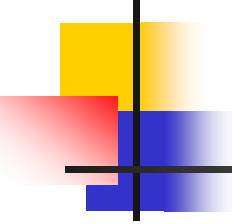
- Review Scenario 5-6 (in the text) involving Lee, a (hypothetical) 35-year old executive named Lee, who:
  - applies for an automobile loan for a BMW;
  - has an impeccable credit history.
- A data-mining algorithm “discovers” that:
  - I. Lee belongs to a group of individuals likely to start their own business;
  - II. people who start business in this field are also likely to declare bankruptcy within the first three years;
- Lee is denied the loan for the BMW based on the profile revealed by the data-mining algorithms, despite his credit score.



# Data Mining (Continued)

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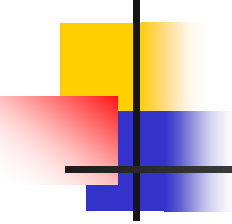
- Although the preceding scenario (involving Lee) is merely hypothetical, an actual case (that was similar to this) occurred in 2008.
- In that incident, a person had two credit cards revoked and had the limit on a third credit card reduced because of certain associations that the company made with respect to *where* this person:
  - shopped,
  - lived,
  - did his banking (Stuckey 2009).



# Data Mining (Continued)

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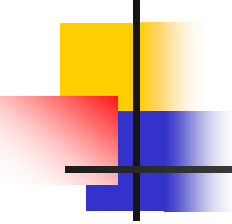
- In that (2008) case, a data-mining algorithm used by the bank “discovered” that this person (whose credit cards were revoked):
  - purchased goods at a store where typical patrons who also purchased items there defaulted on their credited card payments;
  - lived in an area that had a high rate of home foreclosures, even though he made his mortgage payments on time.



## *Web Mining: Data Mining on the Web*

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- Traditionally, most data mining was done in large “data warehouses” (i.e., off-line).
- Data mining is now also used by commercial Web sites to analyze data about Internet users, which can then be sold to third parties.
- This process is sometimes referred to as “Web mining.”
- Review Scenario 5-7 (in the text) on “Facebook Beacon,” as an example of Web mining.



## Table 5-2: Three Techniques Used to Manipulate Personal Data

|                      |                                                                                                                                                                                                                               |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Data Merging</b>  | A data-exchanging process in which personal data from two or more sources is combined to create a "mosaic" of individuals that would not be discernable from the individual pieces of data alone.                             |
| <b>Data Matching</b> | A technique in which two or more unrelated pieces of personal information are cross-referenced and compared to generate a match or "hit," that suggests a person's connection with two or more groups.                        |
| <b>Data Mining</b>   | A technique for "unearthing" implicit patterns in large databases or "data warehouses," revealing statistical data that associates individuals with non-obvious groups; user profiles can be constructed from these patterns. |